Week 4 HW Question 17:

$4x_1 - x_2 + x_3 = -11$

$2x_1 + 2x_2 + 3x_3 = 7$

$5x_1 - 2x_2 + 6x_3 = -12$

$$A = \begin{bmatrix} 4 & -1 & 1 \\ 2 & 2 & 3 \\ 5 & -2 & 6 \end{bmatrix} \quad b = \begin{bmatrix} -11 \\ 7 \\ -12 \end{bmatrix}$$

1.) $\det(A) = 4\begin{vmatrix} 2 & 3 \\ -2 & 6 \end{vmatrix} - (-1)\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ 5 & -2 \end{vmatrix}$

$= 4(12+6) + 1(12-15) + 1(-4-10) = 55$

2.) $A_1 = \begin{bmatrix} -11 & -1 & 1 \\ 7 & 2 & 3 \\ -12 & -2 & 6 \end{bmatrix}$ $\det(A_1) = (-11)\begin{vmatrix} 2 & 3 \\ -2 & 6 \end{vmatrix} - (-1)\begin{vmatrix} 7 & 3 \\ -12 & 6 \end{vmatrix} + 1\begin{vmatrix} 7 & 2 \\ -12 & -2 \end{vmatrix}$

$= (-11)(18) - (-1)(78) + 1(10) = -110$

3.) $A_2 = \begin{bmatrix} 4 & -11 & 1 \\ 2 & 7 & 3 \\ 5 & -12 & 6 \end{bmatrix}$ $\det(A_2) = 4\begin{vmatrix} 7 & 3 \\ -12 & 6 \end{vmatrix} - (-11)\begin{vmatrix} 2 & 3 \\ 5 & 6 \end{vmatrix} + 1\begin{vmatrix} 2 & 7 \\ 5 & -12 \end{vmatrix}$

$= 4(78) - (-11)(-3) + (1)(-59) = 220$

4.) $A_3 = \begin{bmatrix} 4 & -1 & -11 \\ 2 & 2 & 7 \\ 5 & -2 & -12 \end{bmatrix}$ $\det(A_3) = 4\begin{vmatrix} 2 & 7 \\ -2 & -12 \end{vmatrix} - (-1)\begin{vmatrix} 2 & 7 \\ 5 & -12 \end{vmatrix} + (-11)\begin{vmatrix} 2 & 2 \\ 5 & -2 \end{vmatrix}$

$= 4(-10) - (-1)(-59) + (-11)(-14) = 55$

5.) $x_1 = \dfrac{\det(A_1)}{\det(A)} = \dfrac{-110}{55} = -2$ $\quad x_2 = \dfrac{\det(A_2)}{\det(A)} = \dfrac{220}{55} = 4$

$x_3 = \dfrac{\det(A_3)}{\det(A)} = \dfrac{55}{55} = 1$ $\quad (x_1, x_2, x_3) = (-2, 4, 1)$